

Day 7: Power Query + M Language Mastery

[DEF] Definition to Remember

M (Mashup Language) is the functional, case-sensitive programming language that powers all Power Query ETL operations behind the graphical user interface.

1. Structure of an M Query: `let ... in`

Every M expression is built around a let block (defining step variables) and an in block (returning final output):

```
let
    // Step 1: Connect to Source
    Source = Csv.Document(File.Contents("C:\Data\Sales.csv"), [Delimiter=",", Columns=4, Encoding=65001]),

    // Step 2: Promote Headers
    PromotedHeaders = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),

    // Step 3: Change Column Types
    ChangedTypes = Table.TransformColumnTypes(PromotedHeaders, {
        {"OrderID", Int64.Type},
        {"SalesAmount", type number},
        {"OrderDate", type date}
    }),

    // Step 4: Filter Rows
    FilteredRows = Table.SelectRows(ChangedTypes, each [SalesAmount] > 0)
in
    FilteredRows
```

2. Core M Language Concepts

1. Case Sensitivity: Function names and column identifiers are strictly case-sensitive (Table.SelectRows != table.selectrows).

2. Im

3. Writing Custom Reusable M Functions

Converting a sequence of cleaning steps into a parameterized reusable function:

```
(input
let
    Fi

in
    Fi
```

[MUST-WRITE] Must-Master Skills & Interview Points

1. Understand the let ... in syntax hierarchy.
2. How to use Advanced Editor to debug broken ETL steps and parameterize file paths.
3. Creating custom M functions for automated date and text parsing.

[RECALL] Quick Recall

let (step definitions) in (output result) -- Case-sensitive, functional, non-destructive